

PROJECT AND TEAM INFORMATION

Project Title

Smart Campus Lost and Found Matching System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Students and faculty often lose many important things like id cards, wallet, phone, keys, books bags, earphones etc in the campus. Mostly these lost items are found by someone but not sure who exactly lost them, hence it becomes hard to return them to their respective owner. Currently the lost found items are shared on WhatsApp class groups, notice boards or just asking around, making it a bit chaotic since losing track of what was given and who lost them and also there is no search feature. The purpose of our Smart Campus Lost and Found system is to give an overview of what is being found and by whom and where the item was found and also a feature to search for a lost item and notify if found or not. It is crucial to make this a very useful and important project so it is easier to have lost item returned to them. This project will also enable us to test our knowledge learnt in Data Structures using C and also Object Oriented Programming using C++.

State of the Art / Current solution

At Present the lost and found items in campus mainly relies upon WhatsApp groups, Instagram confession page, class groups or by asking to friends and staff. Usually when items get lost inside college campus or outside near college campus, students either report the item to their class group or report to college office or. For people who lost their items they have very limited options, either they messaged on whatsapp class groups, Instagram confession pages or just asking and looking around by themselves to see if anyone has found thier stuff. However currently the colleges dont have a proper system to handle out lost and found items, while the current approach may work for sometimes but it can often be time consuming and complicated. So to make things easier and tackle this problem we created a project where it creates just a one single platform which makes it easier and less time consuming for the user.

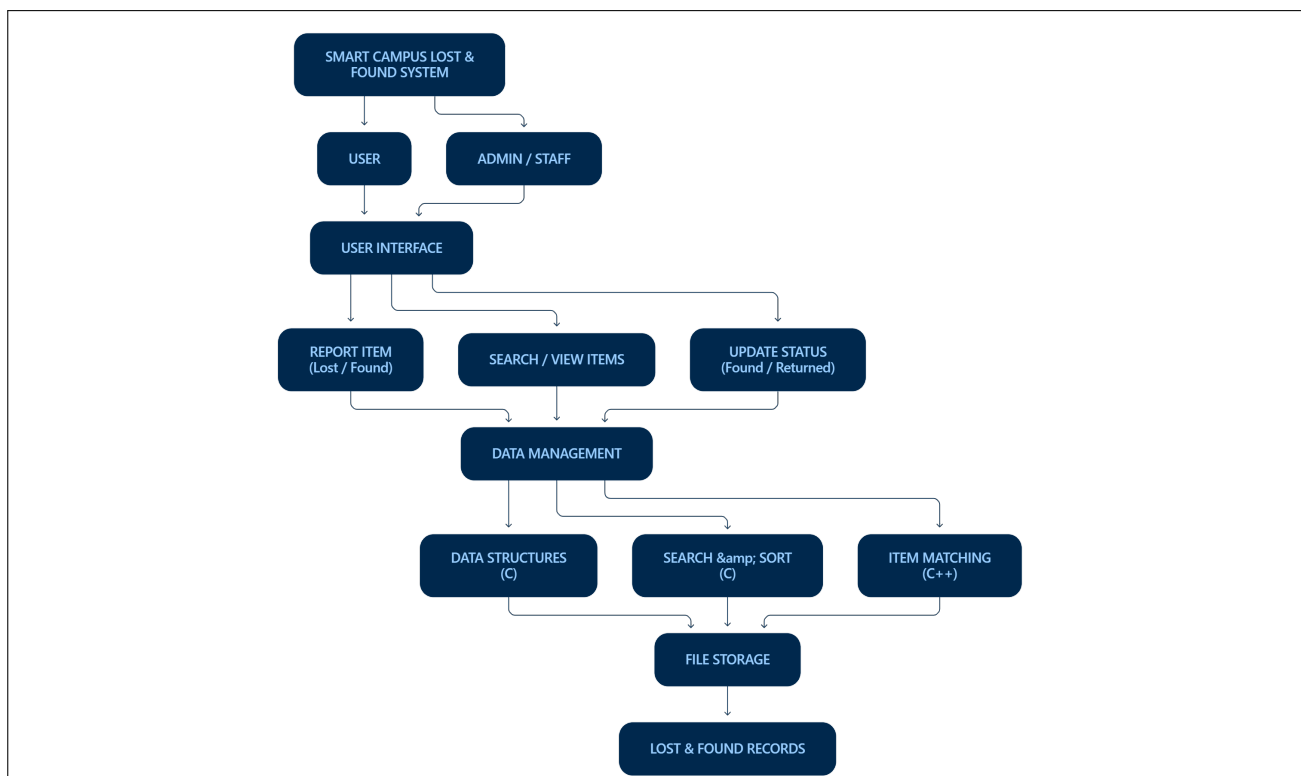
Project Goals and Milestones

The main objective of our project is to design a Smart Campus Lost and Found System that will help students and staff find lost items more efficiently. This will enable users to report lost and found items using item name, location, date, and additional information. It will also include searching options, item matching options, and status updates on located items. Our initial milestones focus on pinpointing the existing processes and challenges within the lost and found system. To tackle this, we plan to utilize Data Structures in C for designing and implementing a storage management system. Additionally, we'll incorporate OOP concepts in C++ to bring our proposed system model to life. As we move forward, our next milestones will involve developing algorithms for searching, sorting, and item matching, followed by thorough testing and debugging of these algorithms to enhance system functionality. Ultimately, our project aims to streamline the lost and found system on campus, making it quicker and more efficient for everyone involved.

Project Approach

We intend to develop the Smart Campus Lost and Found System as a simple application where students and staff report, search, match, and update lost and found items. Firstly, we will determine the features and the general flow of the system. We will develop Data Structures and concepts like searching, sorting, linked lists, stacks, and queues using C. We will apply C++ to develop OOP concepts such as classes, objects, inheritance, and encapsulation. Initially, the project will be developed as a console-based application using either the Visual Studio Code. We will also utilize file handling to store records and test the system under different scenarios to enhance its efficiency.

System Architecture



Project Outcome / Deliverables

The main outcome of our project will be a working Smart Campus Lost and Found System that will simply provide an organized way to manage lost and found items on campus. Users will be able to add details of lost or found items, search for items, search for possible matches, and update the status on the return of an item. This will also demonstrate the use of Data Structures in C such as linked lists, stacks, queues, searching, and sorting, as well as OOP concepts in C++ such as classes, objects, inheritance, and encapsulation. The final model will include the completed working application, file-based data And finally it should make it easier and faster for students and staff regarding storage, testing results, and project documentation to report and retrieve lost items.

Assumptions

- 1. We are assuming that the students and staff will provide accurate and useful information when they submit it. lost or found*
- 2. It is assumed that the user will have available a computer or other device on which the application can be used.*
- 3. We are assuming that the program will be mostly used on the college campus, where the amount stored in the records will be small and therefore will not put a strain on the data structures.*
- 4. It is assumed that the user will update the status of the item when it has been found or returned, and will provide sufficient information so that the item get back returned to its owner.*

References

We referred to the following resources while planning and developing the project:

- 1. GeeksforGeeks – C Programming, Data Structures and C++: <https://www.geeksforgeeks.org/>*
 - 2. W3Schools – C and C++ basics: <https://www.w3schools.com/>*
 - 3. Programiz – C and C++ programming concepts: <https://www.programiz.com/>*
 - 4. C++ Reference – C++ language and OOP concepts: <https://en.cppreference.com/>*
 - 5. TutorialsPoint – Data Structures and C++ programming: <https://www.tutorialspoint.com/>*
- We also referred to our college lecture notes and class materials for Data Structures in C and Object-Oriented Programming in C++.*